Year 11 Higher

REVISION PACK

Handling Data 4

Name: _____

Class: _____

Handling Data 4 · Collecting data and averages (Lessons 4.1, 4.5)

Worked example: questionnaires

Ali uses this question on a questionnaire to find out how much students spend on snacks. Write down two things that are wrong with the question.

How much do you spend on snacks?
0–1 KD ☐ 1–2 KD ☐ 2–3 KD ☐

Step 1	Is there a time?	No: per day or per week?
Step 2	Do boxes overlap?	Yes: 1 KD is in two boxes
Step 3	Is every answer there?	No box for more than 3 KD

Practice: questionnaires

1.1 Huda uses this question on a questionnaire to find out how much time students spend on homework. Write down two things that are wrong with the question.

How many hours of homework do you do?
0–2 ☐ 2–4 ☐ 4–6 ☐

1.2 Majed uses this question on a questionnaire to find out how often people exercise. Write down two things that are wrong with the question.

How many times do you exercise?
1–3 ☐ 3–5 ☐ 5–7 ☐

1.3 Layla uses this question on a questionnaire to find out how much people spend on lunch. Write down two things that are wrong with the question.

How much do you spend on lunch?
0–2 KD ☐ 2–4 KD ☐ 4–6 KD ☐

Practice: questionnaires (continued)

1.4 Omar uses this question on a questionnaire to find out how much time people spend on their phones. Write down two things that are wrong with the question.

> How many hours do you spend on your phone?
> 0–2 ☐ 2–4 ☐ 4–6 ☐

1.5 Mona uses this question on a questionnaire to find out how much people spend on clothes. Write down two things that are wrong with the question.

> How much do you spend on clothes?
> 0–10 KD ☐ 10–20 KD ☐ 20–30 KD ☐

1.6 Fahad uses this question on a questionnaire to find out how many books students read. Write down two things that are wrong with the question.

> How many books do you read?
> 1–2 ☐ 2–3 ☐ 3–4 ☐

Handling Data 4 · Collecting data and averages (Lessons 4.1, 4.5)

Worked example: averages from a frequency table

The table shows the numbers of goals scored by a team in 20 matches.

Number of goals	0	1	2	3
Frequency	5	8	4	3

(a) Write down the mode. (b) Work out the mean number of goals.

(a)	Step 1	Highest frequency	8
	Step 2	Answer	1
(b)	Step 1	Goals × frequency	$0 + 8 + 8 + 9 = 25$
	Step 2	Total frequency	$5 + 8 + 4 + 3 = 20$
	Step 3	Mean	$25 \div 20 = 1.25$

Practice: averages from a frequency table

1.7 The table shows the numbers of goals scored by a team in 25 matches.

Number of goals	0	1	2	3	4
Frequency	3	7	8	5	2

(a) Write down the mode. (b) Work out the mean.

1.8 The table shows the numbers of emails Sara got on 25 days.

Number of emails	1	2	3	4	5
Frequency	2	9	6	4	4

(a) Write down the mode. (b) Work out the mean.

1.9 The table shows the numbers of visitors to a shop each hour for 25 hours.

Number of visitors	0	1	2	3	4
Frequency	6	5	4	8	2

(a) Write down the mode. (b) Work out the mean.

1.10 The table shows the numbers of pets owned by 20 students.

Number of pets	0	1	2	3	4
Frequency	5	9	4	1	1

(a) Write down the mode. (b) Work out the mean.

1.11 The table shows the numbers of books read by 20 students.

Number of books	0	1	2	3	4
Frequency	2	4	8	4	2

(a) Write down the mode. (b) Work out the mean.

1.12 The table shows the numbers of brothers and sisters of 20 students.

Number of brothers and sisters	0	1	2	3	4
Frequency	3	8	6	2	1

(a) Write down the mode. (b) Work out the mean.

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Worked example: a missing number from the mean

The mean of four numbers is 9. Three of the numbers are 7, 8 and 12. Work out the fourth number.

Step 1	Total	$4 \times 9 = 36$
Step 2	Add the three	$7 + 8 + 12 = 27$
Step 3	Subtract	$36 - 27 = 9$

Practice: a missing number from the mean

1.13 The mean of five numbers is 16. Four of the numbers are 12, 14, 19 and 20. Work out the fifth number.

1.14 The mean of six numbers is 11. Five of the numbers are 8, 9, 13, 14 and 10. Work out the sixth number.

1.15 The mean of five numbers is 23. Four of the numbers are 18, 25, 27 and 20. Work out the fifth number.

1.16 The mean of five numbers is 12. Four of the numbers are 9, 11, 14 and 15. Work out the fifth number.

1.17 The mean of four numbers is 20. Three of the numbers are 18, 25 and 16. Work out the fourth number.

1.18 The mean of six numbers is 15. Five of the numbers are 12, 18, 14, 20 and 11. Work out the sixth number.